How Chronos-Bolt makes a forecast

Amazon · 48M parameters · T5 encoder-decoder · zero-shot

1. One series at a time

Chronos-Bolt reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

Covariates such as day length are not used.

Five metal prices become five separate forecasts.

2. Scale, then cut into patches

The context is scaled by its own mean μ and standard deviation σ.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value seen?

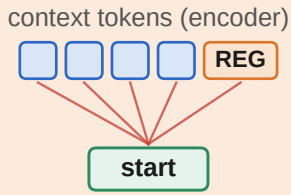
one patch = 16 steps

residual MLP

one token: 512 numbers

3. Encoder reads, decoder writes

Each layer starts with RMSNorm and has a shortcut around it. A REG token closes the context.



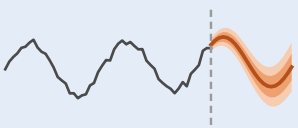
Encoder, 6 blocks: self-attention over all context patches (T5 relative positions)

Decoder, 6 blocks: one start token reads the encoder through cross-attention

Feed forward: 512 → 2,048 → 512, ReLU

4. 64 steps at once

An output network turns the decoder token into **9 quantiles** (0.1 to 0.9) for each of **64 steps**.



median

60% band

80% band

$x = \mu + \sigma \cdot \tilde{x}$ restores the units; the median is the point forecast.

Past 64 steps, each of the 9 quantile paths is fed back in and the 81 results are cut back to 9 quantiles.

A single decoder pass writes 64 steps, so Chronos-Bolt is one of the fastest models here.

6 encoder + 6 decoder blocks · width 512 · 8 heads · context up to 2,048 steps · 64 steps per pass

In the notebooks: chronos-bolt-small · last 2,048 steps · no covariates · each target channel on its own · 90-day horizon takes two passes